

DongJae Seo

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🔗 <https://dongjaeseo.github.io>

Education

Indian Institute of Technology Madras, Aug 2016 – May 2024
B.Tech in Engineering Design

Indian Institute of Technology Madras, Aug 2016 – May 2024
M.Tech(Dual Degree) in Automotive Engineering

Research Experience

Seoul National University, AIoT Lab, Lab Intern Jul 2025 – Present

- Implemented baseline experiments for sleep staging using the MOMENT time-series foundation model.
- Designed preprocessing approaches to adapt long PSG signals for transformer-based models.
- Assessed embedding quality through linear/SVM probing; analyzed performance trade-offs across classification heads.

IIT Madras, Medical Imaging & Reconstruction Lab, Dec 2023 – May 2024
Lab Intern

Chronic Pancreatitis Prognosis Using CT-Based Radiomics

- Built a 3D pancreas segmentation model using SwinUNetR (Dice score: 0.7833).
- Applied post-processing techniques (e.g., connected component analysis) to improve segmentation quality in low-contrast regions.
- Collaborated with SRM Hospital to integrate segmentation results with calcification data.
- Supported radiomic feature extraction pipeline for predicting pancreatitis prognosis.

Professional Experience

Inuaid Solutions, Intern 🌐 Jan 2023 – Jul 2023

- Developed preprocessing pipeline for detecting small nodules in lung CT scans.
- Worked on nodule candidate generation and classification (malignant vs benign) using radiomic features.
- Contributed to the development and integration of ML modules for the diagnostic pipeline.

Relevant Coursework

Multivariate Calculus, Linear Algebra, Design and Analysis of Algorithms, Computer Systems Design, Game Theory, Deep Learning for Imaging, Programming in Python

Course Projects

- *Battery Life Prediction using AI* – Energy Storage Devices and Systems
Developed regression model to estimate battery life using time-series data and performance history.
- *Deep Learning Model Implementation* – Deep Learning for Imaging
Built a complete deep learning training pipeline from scratch using Python and NumPy.
Implemented core components such as forward/backward propagation, weight initialization, CNN/RNN architectures, and optimizers including Adam and SGD, without using high-level libraries.